PROJECT PROPOSAL

Adaptive Control of Solar Power Shufflers for Maximum Power Point Tracking Under Partial Shading

APPLICATION OF MODERN STATE-SPACE METHODS
TO SERIAL PARTIAL POWER CONVERTERS

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Version	Date	Author	Changes / Comments
1.0	17/04/26	Jared Saunders	Initial Proposal

1 General Objective

Photovoltaic (PV) arrays are the rising star of the renewable energy industry. PV arrays are ubiquitous within modern society ranging from small scale outdoor lighting to expansive grid-connected solar farms. Regardless of the scale, PV arrays are subject to the problem of shade, reducing and/or preventing irradiance from being captured over a portion of the array area. Partial shading affects the electrical characteristics of the PV system, subjecting the arrays to drops in performance and introducing potential for hotspot formation that can permanently damage the system. Partial shading is as unavoidable and immutable as the sun rising in the morning, however, the effects can be mitigated. Novel power converter architectures, and maximum power point tracking (MPPT) techniques have been designed for this purpose. This project is scoped to exploring control systems for serial partial power converter (S-PPC) or as known in this report as “power shufflers” [1]. Power shufflers process only the differential power between adjacent modules by redistributing current to effectively equalise the voltage across each module in a string. This ensures all the PV cells remain in operating conditions. The core research question driving this project is; how can power shufflers be controlled to reliably extract the maximum power point of a string under partial shading conditions, whilst having fast convergence to the MPP and maintaining a stable power output? The objective of this project is to design and implement a universal modern control system that enables the use of power shufflers for any PV string configuration and to undergo an experiment that evaluates the ability of the proposed system to track the true maximum power point of partially shaded PV strings.

2 Key Finding from the Literature - Impact on Your Project Design

The literature review identifies four themes of direct relevance to this project’s design. First, partial shading is the dominant cause of PV underperformance in practice, introducing multiple local maxima in the power curve that defeat conventional MPPT algorithms [11], [13], [29]. Second, static hardware mitigations bypass diodes, array reconfiguration, and cross-tied configurations reduce mismatch losses but cannot eliminate them [7], [8], [20]. Third, serial partial power converters (S-PPCs) represent the most efficient hardware architecture for module-level mismatch correction: because they process only the differential mismatch power rather than full string power, converter ratings and losses scale with the degree of shading rather than total array output [1], [6], [12], [18]. Fourth, the control systems literature for S-PPCs is sparse. Existing MPPT work focuses on conventional boost-converter architectures and does not address the MIMO dynamics introduced by a shuffler string. The most relevant state-space formulation [10] models a two-stage PV–boost–MPPT system but does not include shufflers. Modern optimal control methods applied to PV systems LQR, MPC, Kalman filtering demonstrate superior tracking performance over classical P&O [16], [21], [27], but none have been applied to the S-PPC topology. This gap the absence of a modern state-space control framework designed specifically for a shuffler string is precisely what this project addresses. The full literature review can be found in Appendix A.

3 Stakeholders and Resources

The design and implementation of the power shufflers is a team effort. Son Nguyen has taken a hardware focus and is responsible for researching, designing and implementing the electronics. I am taking a control systems focus and am responsible for researching MPPT control techniques, designing the firmware for operating the power shufflers and ensuring the system can safely interface with other components. Our supervisor Geoff Walker will be kept apprised of project progress in-person on a weekly basis. The outcomes of this project add to partial shading, PV panel and S-PPC control systems research. Ideally having a positive impact though physically testing and integrating state of the art theories. The renewable energy industry in turn benefits from future PV system developments.

Table 1: Project Resources

Category	Resource / Consideration
Hardware	PV modules — 2 required per single shuffler validation, 3 for a pair, 4+ for a string; 12+ modules recommended for multi-string experiments.
	Microcontrollers for digital signal processing and closed-loop control.
	Shufflers — one per inter-module interval (number of modules minus one per experiment).
	MPPTs — one unit per string.
	Solar irradiance meter for experimental input characterisation.
	Voltage source and/or resistive load to emulate an energy storage system.
	Raspberry Pi 4B for data acquisition, processing, and PC transmission.
Software	C — primary firmware and simulation language; low-level, safety-critical emphasis.
	MATLAB / Python — secondary languages for data analysis and post-processing.
	MATLAB / Simulink — control design environment with extensive existing toolboxes.
	QSPICE — hardware-in-the-loop (HIL) simulation environment.
	VS Code with PlatformIO extension — firmware build and deployment pipeline.
	Logging application for historical and live experimental data records.
	Facebook Messenger — day-to-day team communication.
	Microsoft Outlook — formal team and supervisor correspondence.
	Microsoft OneDrive / GitHub — file sharing and collaborative version control.
Data	Datasheets for all electronic components used in the design.
	Standard PV array configuration references (series, parallel, series-parallel).
	Representative irradiance and temperature profiles for realistic PV operating conditions.
	Experimental datasets: operating points of PV strings, individual modules, shufflers, and MPPTs.
	Control performance metrics: MPP convergence speed, power overshoot, and steady-state error.
Funding	Shuffler and MPPT manufacturing budget: $500.
	Experiment setup budget: $500.
	Source: founder capital (personal investment) with QUT electronics store concessions.
Ethics & Safety	Compliance with QUT HSE and electrical workspace safety protocols.
	High-voltage safety — experiments involve up to 50 V DC; appropriate isolation and PPE required.
	MISRA-C coding standard for safety-critical embedded firmware.
	If machine learning (e.g. RL for control benchmarking) is adopted: explainability, bias assessment, and model transparency obligations apply.
	Intellectual property awareness — monitor IP Australia obligations throughout development.

4 Project Methodology

The project management approach is a hybrid of the V-model and critical chain scheduling. The V-model defines the full product lifecycle from requirements capture through to system validation while critical chain analysis provides the official timeline by accounting for resource constraints and buffers. The critical path defines the logical task sequence within that structure. This hybrid framework was chosen because the V-model alone does not address team coordination, task ownership, or funding management. Its core strength is traceability: requirements defined early are directly linked to test plans and acceptance criteria at every subsequent design level.

Methods

1. **MATLAB Simulation.** MATLAB is the primary environment for control systems design. It supports rapid prototyping and iteration of control architectures, including state-space modelling, stability analysis, linearisation, discretisation, and controller synthesis. Simulink extends this to block-diagram simulation of the closed-loop system prior to hardware implementation.

2. **SPICE Simulation (QSpice).** QSpice is used for hardware-level circuit simulation, applying the designed control system to the power electronics model of the shuffler. This allows verification of hardware-specific behaviour such as switching dynamics, operating point shifts under varying irradiance, and transient response all without physical hardware.

3. **PCB Design (Kicad).** The power shuffler and custom MPPT are realised as printed circuit boards designed in Altium Designer. This method bridges the gap between simulation and physical experiment, producing manufacturable hardware for bench validation.

4. **Firmware Development (VS Code / PlatformIO).** Embedded firmware for the microcontroller-based control system is developed in C using VS Code with the PlatformIO extension. This may also include companion software for live data visualisation and logging from the embedded system to a personal computer.

5. **Physical Experiment.** The complete shuffling system is validated on a bench test rig comprising one or more PV panels, the embedded control system, a bench harness, and a high-voltage DC load emulating an energy storage system. Experimental results are compared against simulation predictions to assess tracking performance, convergence speed, and steady-state error.

6. **Machine Learning Validation (Supplementary).** If time permits, a reinforcement learning agent will be developed as an independent benchmark for the designed control system. The RL model provides a data-driven upper bound on achievable performance, against which the analytical controller can be evaluated.

Control Systems Design Process

The power shuffler is a multiple-input multiple-output (MIMO) system whose power output characterised by voltage and current is a nonlinear function of irradiance and temperature. The control design process follows modern state-space methods and proceeds in the following sequence:

1. **System modelling.** Derive ordinary differential equations (ODEs) governing the dynamics of the PV panels, shuffler power circuit, and MPPT algorithm. Express the system in continuous-time state-space form.

2. **System analysis.** Assess stability, controllability, and observability of the continuous-time model. Identify equilibrium operating points under representative irradiance and temperature conditions.

3. **Linearisation.** Linearise the nonlinear system about each equilibrium point to obtain a family of linear time-invariant (LTI) models suitable for classical and modern controller design.

4. **Discretisation.** Convert the continuous-time model to discrete time, selecting an appropriate sampling period that satisfies Nyquist constraints and accounts for microcontroller timing.

5. **State feedback control.** Design a state feedback controller addressing both regulation (disturbance rejection) and reference tracking. Evaluate optimal control formulations (e.g. LQR) to minimise a cost function over the operating envelope.

6. **State estimation.** Implement output feedback via a state observer (Luenberger) or a Kalman filter where measurement noise is significant, enabling full-state feedback from available sensor outputs only.

7. **Advanced methods (as applicable).** Consider emulation-based design (eg. Zero Order Hold, Bilinear Transform, Pole-Zero Matching), gain scheduling across operating points, or model predictive control (MPC) for constraint-aware optimisation of the MPPT trajectory.

5 Deliverables

To achieve the general objective for the project the primary deliverables to be produced by project hand-off include: A physical power shuffler and maximum power point tracker prototype. A demonstration video of prototype during an experiment. Presentation of findings and results of the project. A final project report documenting the important developments of the project and evaluating the outcomes.

The artefacts produced during the design phase will include: At least three iterations of the power shuffler. At least two iterations of the custom maximum power point tracker. Each iteration will have a build of materials, a schematic, test results and a design review. The design of the control system will begin in MATLAB / SIMULINK for ease of design due to familiarity. QSPICE files simulating firmware running on the hardware.

During the early research and planning phases of the project they key artefact is a literature review of existing control systems solutions to the partial shading problem. This document will be included as the project plan.

Public access to a GitHub repository collating not just firmware source code but all the digital files we used including MATLAB control models, QSPICE files, datasets, PCB schematics, and further documentation. The GitHub itself will have handy user manuals in the form of 'README' files to assist in understanding how to navigate the repository itself.

6 Risks, Requirements and Constraints

The project interfaces with existing electrical infrastructure (QUT laboratory power supplies, PV test rigs) and is subject to QUT HSE electrical workspace regulations and MISRA-C coding standards for safety-critical firmware. Scope is constrained by the dependency on power electronics hardware delivered by the second team member control systems work that requires a physical shuffler cannot proceed until hardware milestones are met. The full risk register and system requirements table can be found in Appendix C. This is a summary of the most important risks, and constraints.

The most critical risk is the electrical hazard 50 V DC is genuinely dangerous and is the only item rated Critical, demanding strict QUT HSE compliance and mandatory PPE. Two High-rated risks are specific to control systems: firmware failure and control-hardware integration, both mitigated through the HIL simulation pipeline before any hardware is touched. MPPT global tracking failure is also flagged as a control systems-specific High risk the algorithm must be proven across all partial shading scenarios in simulation before deployment, since a local-maximum trap would invalidate the entire experimental result. Scope creep and assignment deadline collision are the dominant project management risks, both rated High and addressed through the one-month contingency buffer and change control process. Team member withdrawal is a Medium risk but has asymmetric impact it blocks hardware progress entirely while control systems work can continue in isolation, which is reflected in the mitigation.

The control systems requirements are the most demanding subset. Global MPP tracking is the hardest constraint the system must distinguish and converge to the global maximum rather than any local peak, which is a non-trivial algorithmic requirement under partial shading. The three-tier control hierarchy (individual shuffler, string, multi-string) defines the scalability architecture and directly shapes the firmware structure. MISRA-C compliance is a non-negotiable constraint that affects every line of embedded code written. On the power electronics side, the 10 kHz minimum switching speed and non-isolated topology are the binding hardware constraints that the control system must be designed around. The five partial shading scenarios function as the acceptance test matrix — your system is only considered validated once it performs correctly across all five.

7 Quality and Sustainability

Quality

The quality of deliverables is measured by experimental results structured through formal test plans. Each test plan specifies a test objective, a series of test cases reflecting realistic operating scenarios including edge cases, and the resources required to execute the experiment. This structure ensures that quality assessment is repeatable, traceable, and directly tied to the acceptance criteria defined during the design phase.

Quality is further enforced through the V-model's inherent test traceability structure. Every requirement defined during the design phase has a corresponding acceptance criterion and test plan at the appropriate verification level. Quality is assessed across three escalating scopes:

1. **Unit testing.** Individual components the shuffler circuit, microcontroller firmware modules, and MPPT algorithm are tested in isolation against their specified acceptance criteria before integration. Firmware is validated in simulation (QSpice HIL) prior to deployment on hardware.

2. **Subsystem verification.** The shuffler and MPPT are tested together as a subsystem. Key metrics include MPP convergence speed, power overshoot, and steady-state tracking error under controlled irradiance and temperature conditions. Results are compared against simulation predictions to assess model fidelity.

3. **System validation.** The complete bench rig PV panels, shuffler string, embedded controller, and load is tested as an integrated system. Validation criteria include energy yield improvement relative to a conventional series string under partial shading, and stable closed-loop operation across the expected operating envelope.

Code quality is governed by the MISRA-C standard for safety-critical embedded software, enforcing deterministic behaviour, bounded execution time, and the absence of undefined language constructs. Version control via GitHub provides a full audit trail of firmware changes, supporting reproducibility and peer review.

Sustainability

The power shuffler directly targets **UN Sustainable Development Goal (SDG) 7** *ensure access to affordable, reliable, sustainable and modern energy for all*. The relevant SDG 7 targets addressed by this project are:

- **7.1** By 2030, ensure universal access to affordable, reliable and modern energy services.
- **7.2** By 2030, increase substantially the share of renewable energy in the global energy mix.
- **7.3** By 2030, double the global rate of improvement in energy efficiency.

By recovering energy lost to partial shading mismatch in PV strings, the shuffler directly improves the energy efficiency and output reliability of solar installations contributing to all three targets simultaneously. The sustainability scope of this project is primarily short-to-medium term, focused on embedded energy efficiency and practical deployability. A full life cycle analysis (LCA) is outside the scope of this capstone; however, the following considerations are explicitly accounted for:

1. **Energy efficiency.** The central purpose of the solar power shuffler is to recover energy that would otherwise be lost to partial shading mismatch. Improved MPP tracking directly reduces the levelised cost of energy (LCOE) of a PV installation and extends the effective productive life of an array without additional hardware.

2. **Component selection.** Hardware procurement favours commercially available, RoHS-compliant components. Custom PCB designs minimise board area and layer count to reduce material consumption during manufacturing.

3. **Repairability and modularity.** The shuffler architecture is inherently modular individual units can be replaced or upgraded without replacing the full string. This is a meaningful advantage over conventional panel-level optimisers, which are typically non-serviceable and contribute to e-waste at end of life.

4. **Scalability.** The control system is designed to be topology-agnostic within the S-PPC family, meaning the firmware and algorithms can be redeployed across different string configurations and panel counts without a full redesign. This reduces development overhead for future iterations and supports longer product relevance.

5. **Broader impact.** Partial shading is the dominant real-world cause of PV underperformance in residential and commercial rooftop installations. A cost-effective, manufacturable shuffler control system contributes directly to improving the economic viability of solar adoption, which aligns with Australia's renewable energy transition objectives.

8 Timeline and Deliverables

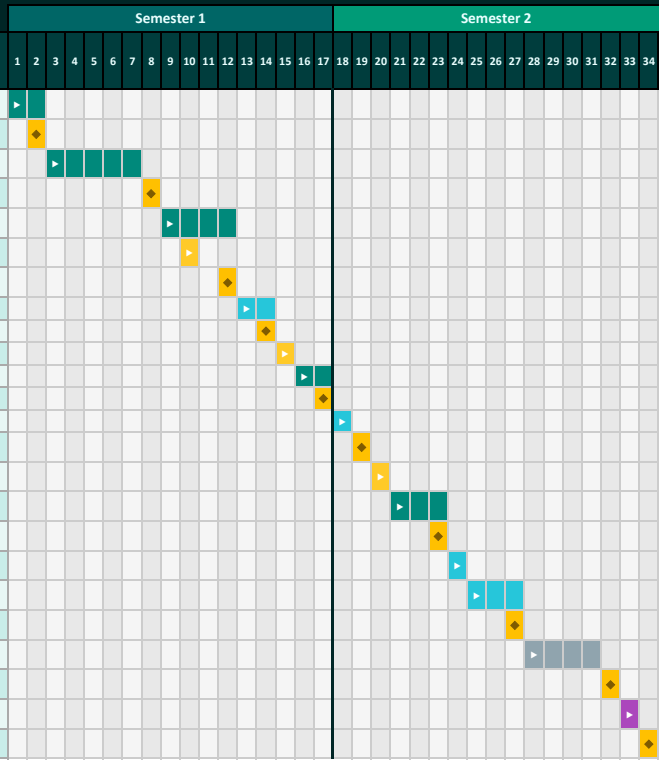
To provide a detailed and visual representation of the timeline for the power shuffler control system a gantt chart was constructed (as seen on the next page). This gantt chart presents the 2 semester project on a 34 week timeline where 1 is semester 1 week 1 and 34 is semester 2 week 14. Break and exam block are included. Each deliverable is assigned a project phase, a set of deliverables, the dependencies between tasks (aka pre-requisites), start date, finish date, and associated milestone. The most important hard deadlines are the printed circuit board orders. Many of the interim delliverables are documentation, designs, experiment reports, test plans and presentations. The development of the power shuffler system relies on the convergence of two parallel projects through the integration of hardware and software subsystems. Appendix B presents the gantt chart for the entire team.

There were 10 key milestones identified. The phases the project goes through follows closely with the V-model as seen in Figure 1. The first milestone is about planning the project, determining how it is intended to be applied in the end user environment, defining system requirements. Milestone 2 moves to high level design by developing an architecture that considers how the components and subsystems will interact. From milestone it becomes iterative in the sense of creating a detailed design, developing it and testing it. For the development of the shuffler there is intended to be at lest three versions. Each version reaching higher up the V. V1 is only unit testing. V2 is susbsytem verification. V3 is System verification. Milestones 9 and 10 capture the close and handoff phase of the project.

1. Literature review completed and project proposal approved
2. System architecture approved
3. Simulation models documented
4. Shuffler V1 testing complete
5. Shuffler V2 and MPPT V1 firmware documented
6. Subsystem verification complete
7. Shuffler V3 and MPPT V2 firmware documented
8. System validated
9. Final report and handover submitted
10. Project presentation

Control Systems Sub-Timeline | Serial-Partial Power Converter

#	Phase	Task / Deliverable	Team	Dependencies	Start Wk	End Wk	MS
1	Setup	Project Setup	Both	—	1	2	
2	Setup	1st Supervisor Meeting & Communication Strategy	Both	T01	2	2	M1
3	Planning	Project Planning & Literature Review	Both	T02	3	7	
4	Design	High-Level System Architecture	Both	T04	8	8	M2
5	Design	Shuffler V1: Circuit Design & Control Model	Control Sys.	T05	9	12	
6	**Order**	**Shuffler V1 Components Ordered**	**Control Sys.**	**T06**	**10**	**10**	
7	Design	Documented Simulation Models	Control Sys.	T06	12	12	M3
8	Testing	Develop Unit Test Plans & Acceptance Criteria	Control Sys.	T06, T07	13	14	
9	Testing	Shuffler V1 Testing & Experiment Report	Control Sys.	T09	14	14	M4
10	**Order**	**Shuffler V2 & MPPT V1 Ordered**	**Control Sys.**	**T08, T10**	**15**	**15**	
11	Design	Shuffler Pair Control (V2) & MPPT V1 Design	Control Sys.	T11	16	17	
12	Design	Documented Shuffler & MPPT Firmware	Control Sys.	T12	17	17	M5
13	Testing	Develop Subsystem Test Plan	Control Sys.	T11	18	18	
14	Testing	Shuffler V2 & MPPT V1 Testing & Reports	Control Sys.	T14	19	19	M6
15	**Order**	**Shuffler V3 & MPPT V2 Ordered**	**Control Sys.**	**T15**	**20**	**20**	
16	Design	Shuffler String Control (V3) & MPPT V2 Design	Control Sys.	T16	21	23	
17	Design	Documented V3 & MPPT V2 Firmware	Control Sys.	T17	23	23	M7
18	Testing	System Test Plan & Risk Assessment	Control Sys.	T16	24	24	
19	Testing	System Validation & Integration Testing	Control Sys.	T19	25	27	
20	Testing	Report Draft & Validated Physical System	Control Sys.	T04, T05, T08, T11, T13, T16, T18, T20	27	27	M8
21	Buffer	Buffer / Contingency	Both	—	28	31	
22	Close	Final Report & Handover Documentation	Both	T21	32	32	M9
23	Close	Presentation Preparation	Both	T32	33	33	
24	Close	Presentation Delivery	Both	T33	34	34	M10



LEGEND
Setup / Planning / Design
Testing / Verification
Order (components/PCB)
Buffer / Contingency
Project Close
◆ Milestone

Notes:

- Work released Mon; due Sun.
- PCBs ordered by Fri morning.
- 3-wk PCB turnaround assumed.
- 1-month integration testing.
- 1-month contingency buffer.

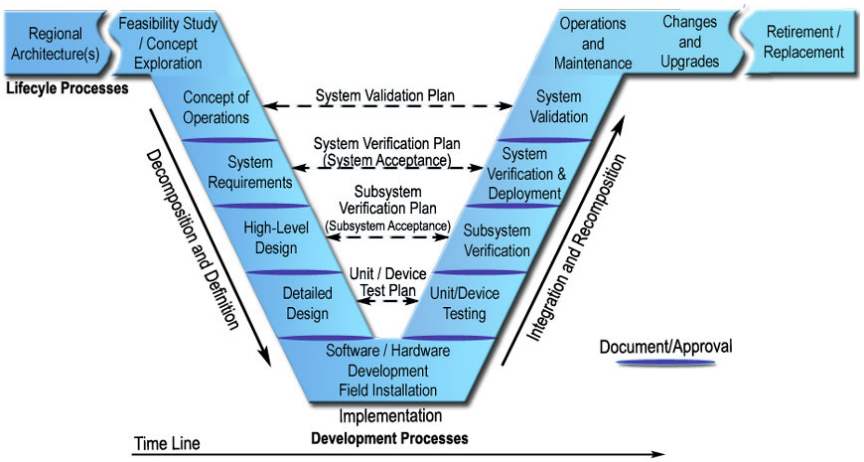


Figure 1: V-model

9 Management of Project Changes

The project timeline is very fluid and is susceptible to project changes. Table 2 provides a brief summary of several key changes and their associated handling strategies. There are four general reasons for changes. Firstly, reductions in scope from project deliverables becoming unfeasible. Secondly, new technical decisions modifying the architecture. Thirdly, sometimes the project management plans are unable to capture all aspects of the project such as complexity and time. Lastly, the competing commitments of the people that make the team causing them to become unreliable.

Table 2: Project Change Management Register

Project Change	Category	Management Strategy
Target Use-case	Scope	The design targets a universal product, so most use-case refinements require only minor adjustments. A domain-specific requirement (e.g. defence-grade robustness) introduces additional constraints that scope to a particular component.
Custom MPPT Design	Scope	Select an off-the-shelf MPPT unit for purchase. If a suitable unit is unavailable or cost-prohibitive, the project will be descoped to treat the shuffler control system as a standalone proof-of-concept without full MPPT integration. This has a significant impact on deliverables for milestones five onwards.
Shuffler Topology	Technical	The firmware is designed around S-PPC. A topology change invalidates the control model and requires a full firmware redesign. This change risks the feasibility of project completion. It is a very large direction requiring large control and hardware redesign. Would require supervisor input before proceeding.
Control System Architecture	Technical	The amount of redesign required depends on the nature of the change. A revised dynamics model requires a full redesign, whereas changes to the operating range (e.g. larger input voltage) constitute minor parameter adjustments only. Very minor impact on shuffler hardware design.
New Requirements	Project	New requirements are captured and deferred to the next design revision of the relevant component, with the associated test plan updated accordingly. Significant requirement changes impact the shuffler / control design as whole would require supervisor input.
Timeline Rearrangements / Extension	Project	A one-month contingency buffer is incorporated into the schedule to handle delays. 1 month was selected for buffer size to accommodate a full PCB reorder cycle. Tasks will be reprioritised using the established dependency structure if the buffer is consumed.
Supervisor Unavailable	People	The project becomes self-directed. Could reach out to other QUT academics in the control systems field for guidance, and all decisions made during the absence will be documented for supervisor review upon return.
Team Member Withdrawal	People	Hardware development would be blocked. Priority shifts to control systems work that can proceed in isolation (simulation, firmware). Physical system tests are deferred until hardware is available.

10 Sign Off

Table 3: Document Sign Off

Student Engineer	Signature	Date
Jared Saunders	*JSaunders*	*18/03/2026*

Appendices

A Review of Literature

A.1 Introduction

Photovoltaic (PV) arrays are becoming more prevalent and have seen a surge of interest by researchers over the past decade. The majority of this research is targeting the one major problem with solar panels; partial shading. There are volumes of literature exploring the impact of partial shading. This review aims to bring together the lessons and perspectives from power electronics and control systems fields on the problem. Often the proposed solutions involve new power circuit topologies with one such power circuit, that remains largely untapped, being a power shuffler, also known as serial partial power converter (S-PPC). The merit of this topology lies in its modularity and the fact that converter ratings scale with mismatch power rather than total array power, making it inherently efficient under realistic shading conditions [1], [12]. The core research question driving this review is: *how can serial partial power converters be controlled to reliably extract the maximum power point of a PV string under partial shading conditions, whilst achieving fast convergence and maintaining stable power output?* This review evaluates the state of the art in MPPT techniques and identifies the design factors a new control system must address. Rather than addressing control systems directly, this review takes a top-down approach. Beginning with an exploration of the renewable energy industry from the lens of photovoltaic arrays, then examining the impact of partial shading on PV performance, existing solutions from research and industry, the S-PPC architecture, and finally the world of PV control systems.

A.2 Solar Energy Industry

Solar photovoltaics is the fastest-growing energy source globally. The IEA World Energy Outlook 2025 confirms that renewables are outpacing all other energy sources across every modelled scenario, led by solar PV [30]. Investment in renewable power has grown consistently since 2015, with solar attracting the largest share of new capacity additions worldwide [30]. Australia is a strong participant in this trend — the Clean Energy Regulator reported 6.8 GW of new renewable capacity added in 2025 alone, with renewables now accounting for 43.7% of National Electricity Market generation [31]. The domestic large-scale solar industry is valued at approximately $2 billion and continues to grow [28], with allied markets in China and the United States operating at a significantly larger scale [25], [26].

PV systems service a broad range of applications: residential rooftops, grid-scale farms, communications infrastructure, aerospace, water pumping, and purpose-built platforms such as solar racing vehicles [22], [24]. Across these applications, two dominant load types emerge grid-tied inverters and energy storage systems such (ESS) as battery packs. The prevalence of energy storage as the typical load case is a recurring theme in the literature and will be a core design consideration, which targets ESS-type loads as the primary operating condition. The niche case of solar racing is particularly relevant to this project's origins with the predecessor of this project attempting to implement a power shuffler on a Team Arrow solar car for the 2019 Bridgestone World Solar Challenge [14], establishing the foundation this work builds upon. Across all applications, the same fundamental problem arises: partial shading.

A.3 Partial Shading

Partial shading occurs whenever cells in a PV array receive reduced irradiance compared to the rest of the array. As the irradiance drops the operating point of the cell moves towards reverse-bias. If the cells

reach the reverse-bias region they effectively acting as open circuits. Because modules are connected in series, the current of the entire string is limited to that of the weakest module. The consequences are twofold: reduced energy yield and the appearance of multiple local maxima in the string power curve, which provides a challenge for MPPT algorithms [11], [13].

What makes partial shading particularly difficult to mitigate is its unpredictability it can occur anywhere and at any time. Sources range from static obstructions such as buildings and trees, to transient events including cloud cover, dust, and soiling, through to structural degradation from cracking or aging [8]. Beyond the performance impact, shading poses a safety risk. Song et al. [29] demonstrate experimentally that partially shaded modules can reach temperatures of 99.6 °C, exceeding the melting point of the ethylene-vinyl acetate encapsulant (EVA, 85 °C) and creating a hotspot fire risk that makes mitigation a safety concern as well as a performance one.

Array configuration affects the degree of mismatch loss. Rao et al. [7] compare series-parallel, bridge-linked, total cross-tied (TCT), dynamic, and pattern-matched configurations, finding TCT offers the best performance among conventional static schemes. Belhaouas et al. [8] and Bingöl & Özkaya [11] extend this comparison across multiple shading arrangements on larger arrays, confirming that series-parallel configurations sacrifice significant power under partial shading. Mehedi et al. [20] provide a systematic review of partial shading mitigation hardware at both the module and array level. The conclusion across all these studies is consistent: bypass diodes reduce mismatch losses but introduce misleading losses by creating local maxima that trap conventional MPPT algorithms, while static reconfiguration is impractical without complete prior knowledge of the installation environment. These limitations motivate the search for a more adaptive architectural solution which is where power shufflers enter the picture.

A.4 Power Shufflers and the S-PPC Architecture

The concept of processing only the *differential* power between mismatched modules, rather than the full string power, has been explored under several aliases and topological variations over the past two decades. Walker et al. [1] first proposed the idea as a per-module maximum power point enabling converter. Kasper et al. [3] later formalised a classification of panel-integrated DC–DC converter concepts, distinguishing between series and parallel connection points. Olalla et al. [5], [6] and Jeong et al. [12] developed the DPP framing, demonstrating analytically and experimentally that differential processing achieves near-ideal energy extraction under mismatch conditions with converter ratings that scale with shading severity rather than total array power. Li & Cobos [18] formalise this efficiency advantage using a VA-area modelling framework. At the more complex end of the spectrum, Schweizer & Soeiro [9] explore multi-level flying capacitor architectures that push switching performance further, and Jeong et al. [15] introduce segmented DPP units with group-level control algorithms.

This body of work spans a wide range of converter complexities and connection configurations. For this project, the focus is deliberately narrowed to the simplest and most practical branch of the DPP family: the serial partial power converter (S-PPC) with a straightforward bidirectional converter topology such as a buck-boost, Ćuk, half-bridge, or full-bridge. The S-PPC connects in series with the string power path, processing module-to-module current differentials to equalise the voltage across each module. This topology is chosen for its simplicity and close resemblance to the Walker shuffler concept.

Dos Santos et al. [17] present a two-switch forward S-PPC capable of step-up/down operation, providing a practical reference topology for the hardware design. Kasper et al. [4] analyse the impact of shading conditions on panel voltage equalising converters, providing circuit-level design constraints relevant to the power electronics implementation. Next the question of how to control the shufflers and MPPT are addressed.

A.5 MPPT Control Systems

The MPPT control literature is extensive but predominantly focused on conventional single-converter architectures. The control problem for this project is a multiple input multiple output system with coupled dynamics across the string remains largely unsolved in the literature. The previous project [14] developed an initial MPPT control algorithm for the shuffler but did not apply modern state-space methods, establishing the baseline this project builds upon. Esram & Chapman [2] provide a survey of MPPT techniques, cataloguing over 19 methods including perturb-and-observe (P&O), incremental conductance, and various intelligent methods. Sarvi & Azadian [24] update this survey, classifying algorithms across classical, bio-inspired, and machine-learning categories. The common limitation across classical methods is susceptibility to local maxima under partial shading; across intelligent methods, the limitations are computational cost and the requirement for large training datasets.

Riquelme-Dominguez & Martinez [23] evaluate P&O variants using state-space models under dynamic test conditions. The key finding is that differential P&O (dP&O) minimises MPP drift compared to standard P&O but does not resolve the global tracking problem.

The most directly relevant contribution to this project's control design is Batzelis et al. [10], who develop a continuous-time state-space representation of the irradiance-driven dynamics of a two-stage PV–boost–MPPT system. The model captures six state variables: boost converter capacitor voltage and inductor current, the PI controller integrator, the MPPT voltage reference, and past voltage and power values. Inputs are irradiance and temperature; the output is PV system power. This provides the foundational modelling methodology for this project, with the addition of the shuffler's bidirectional converter dynamics.

Modern optimal control methods have demonstrated clear performance advantages over classical MPPT. Assabaa et al. [27] implement LQR and LQR-PI controllers for a PV system, finding that LQR-PI outperforms LQR across all metrics including steady-state power and disturbance rejection under irradiance variation. Lin et al. [19] propose a state-space MPPT using reinforcement learning, noting that RL models do not require labelled training data and adapt well to changing environments, but are limited by state-space size. Kacimi et al. [16] implement a hybrid neural network–MPC–Kalman filter MPPT, and Kabalci & Boyar [21] replace the MPPT block entirely with an Extended Kalman Filter on an interleaved boost converter, demonstrating that observer-based control can subsume the tracking function.

None of these modern control designs have been applied to the S-PPC topology. The introduction of shufflers fundamentally changes the system structure: the MIMO plant now includes the dynamics of each shuffler converter, the interactions between adjacent module currents, and the coupling between the shuffler control loop and the MPPT reference trajectory.

A.6 Conclusion and Design Implications

The literature establishes that partial shading is an unpredictable problem that causes energy losses and safety risks across every PV application. When considered at scale across the residential, commercial, and grid-scale installations that collectively define the solar industry the total power being surrendered to partial shading mismatch represents a significant unaddressed opportunity. A practical, scalable mitigation solution would have meaningful impact on the industry as a whole.

Of the hardware architectures reviewed, the S-PPC is the most promising path forward for module-level mismatch correction. Its efficiency advantage is both experimentally demonstrated and theoretically bounded, and its series connection topology is the most practically viable for string-level deployment. This project adopts the S-PPC as its target architecture for a power shuffler.

On the control systems side, the work of Batzelis et al. [10] opens a significant avenue the state-space representation of PV–boost–MPPT dynamics provides a rigorous modelling foundation that this project will extend to incorporate the additional dynamics of the shuffler converters. From that foundation, a range of controller designs become accessible: state feedback, LQR, MPC, and observer-based approaches including Luenberger observers and Kalman filters. The literature demonstrates that each of these has been applied successfully to conventional single-converter PV systems, with modern methods consistently outperforming classical P&O. MATLAB and Simulink are the established tools for this class of control design and will be used throughout.

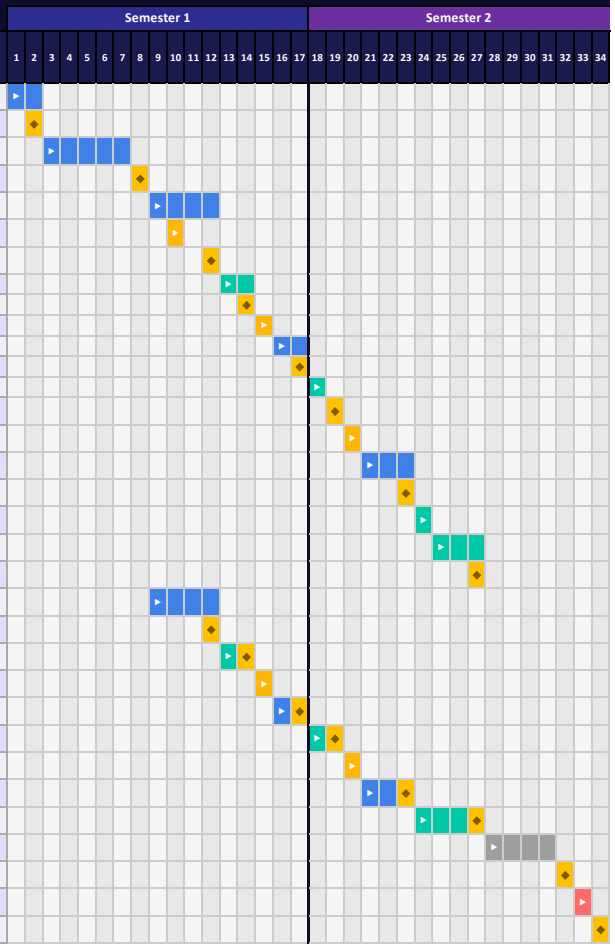
The critical gap is that none of these modern control approaches have been applied to the S-PPC MIMO system. With multiple shufflers operating across a string, the duty cycles of each converter are separate control inputs and the module voltages and currents are coupled outputs adjusting one shuffler shifts the operating point of every adjacent module, changing the conditions under which all others operate. This coupling is what makes the problem genuinely MIMO and what the existing literature has not addressed. Finally, if time permits, a reinforcement learning agent will be developed as an independent benchmark providing a data-driven upper bound on achievable tracking performance against which the analytically designed controllers can be evaluated.

B Team Gantt Chart

(See Next Page)

Solar Power Shuffler — Full Project Timeline | 34 Weeks

#	Phase	Task / Deliverable	Team	Dependencies	Start Wk	End Wk	MS
1	Setup	Project Setup	Both	—	1	2	
2	Setup	1st Supervisor Meeting & Communication Strategy	Both	T01	2	2	M1
3	Planning	Project Planning & Literature Review	Both	T02	3	7	M1
4	Design	High-Level System Architecture	Both	T04	8	8	M2
5	Design	Shuffler V1: Circuit Design & Control Model	Control Sys.	T05	9	12	
6	**Order**	**Shuffler V1 Components Ordered**	**Control Sys.**	**T06**	**10**	**10**	
7	Design	Documented Simulation Models	Control Sys.	T06	12	12	M3
8	Testing	Develop Unit Test Plans & Acceptance Criteria	Control Sys.	T06, T07	13	14	
9	Testing	Shuffler V1 Testing & Experiment Report	Control Sys.	T09	14	14	M4
10	**Order**	**Shuffler V2 & MPPT V1 Ordered**	**Control Sys.**	**T08, T10**	**15**	**15**	
11	Design	Shuffler Pair Control (V2) & MPPT V1 Design	Control Sys.	T11	16	17	
12	Design	Documented Shuffler & MPPT Firmware	Control Sys.	T12	17	17	M5
13	Testing	Develop Subsystem Test Plan	Control Sys.	T11	18	18	
14	Testing	Shuffler V2 & MPPT V1 Testing & Reports	Control Sys.	T14	19	19	M6
15	**Order**	**Shuffler V3 & MPPT V2 Ordered**	**Control Sys.**	**T15**	**20**	**20**	
16	Design	Shuffler String Control (V3) & MPPT V2 Design	Control Sys.	T16	21	23	
17	Design	Documented V3 & MPPT V2 Firmware	Control Sys.	T17	23	23	M7
18	Testing	System Test Plan & Risk Assessment	Control Sys.	T16	24	24	
19	Testing	System Validation & Integration Testing	Control Sys.	T19	25	27	
20	Testing	Report Draft & Validated Physical System	Control Sys.	T04, T05, T08, T11, T13, T16, T18, T20	27	27	M8
21	Design	Shuffler V1: Power Circuit & Hardware Design	Power Elec.	T05	9	12	
22	Design	Hardware Simulation & BOM Documentation	Power Elec.	T22	12	12	M3
23	Testing	Hardware Unit Testing – Shuffler V1	Power Elec.	T22, T07	13	14	M4
24	**Order**	**PCB V2 & MPPT Board Ordered**	**Power Elec.**	**T23, T24**	**15**	**15**	
25	Design	PCB Design: Pair Configuration & MPPT	Power Elec.	T25	16	17	M5
26	Testing	PCB Subsystem Assembly & Verification	Power Elec.	T25, T26	18	19	M6
27	**Order**	**PCB V3 & MPPT V2 Ordered**	**Power Elec.**	**T27**	**20**	**20**	
28	Design	PCB String Configuration Design	Power Elec.	T28	21	23	M7
29	Testing	Full-System Hardware Validation	Power Elec.	T28, T29	24	27	M8
30	Buffer	Buffer / Contingency	Both	—	28	31	
31	Close	Final Report & Handover Documentation	Both	T21	32	32	M9
32	Close	Presentation Preparation	Both	T32	33	33	
33	Close	Presentation Delivery	Both	T33	34	34	M10



LEGEND
Setup / Planning / Design
Testing / Verification
Order (components/PCB)
Buffer / Contingency
Project Close
◆ Milestone

Notes:

- Work released Mon; due Sun.
- PCBs ordered by Fri morning.
- 3-wk PCB turnaround assumed.
- 1-month integration testing.
- 1-month contingency buffer.

C Risks, Requirements and Constraints

The project interfaces with existing electrical infrastructure (QUT laboratory power supplies, PV test rigs) and is subject to QUT HSE electrical workspace regulations and MISRA-C coding standards for safety-critical firmware. Scope is constrained by the dependency on power electronics hardware delivered by the second team member — control systems work that requires a physical shuffler cannot proceed until hardware milestones are met.

Risk Register

Likelihood and consequence are rated Low (L), Medium (M), High (H), or Critical (C). Overall risk rating is the higher of the two scores.

System Requirements and Constraints

Requirements are derived from the project specification and validated against the system design at each V-model level.

Table 4: Project Risk Register

#	Risk	Category	Likelihood	Consequence	Rating	Mitigation / Monitoring Strategy
1	Scope creep	Project	H	H	H	Strict change control; new requirements deferred to next design revision. Supervisor sign-off required for scope additions.
2	No PV hardware available	Resource	M	C	H	Procure panels early in Semester 1. Fallback: validate control system in QSpice HIL simulation and document as a known limitation.
3	Electronics failure	Hardware	M	H	H	Overload protection required by specification. Order spare components. One-month buffer absorbs a PCB reorder cycle.
4	Firmware failure	Control Sys.	M	H	H	MISRA-C enforced. QSpice HIL and MATLAB validate firmware before hardware deployment. Unit test plans with acceptance criteria.
5	Monitoring software inaccuracy	Software	L	M	L	Sensor noise filtering required by specification. Cross-validate logged data against known operating points during calibration.
6	Data loss	Data	L	H	M	GitHub for code; OneDrive auto-sync for datasets. Raw data backed up to secondary local drive after each session.
7	Unable to access data	Data	L	M	L	All assets in shared cloud repositories with at least two admin-level team members.
8	Electrical hazard (50 V DC)	Safety	M	C	**C**	QUT HSE electrical protocols enforced. PPE and isolation procedures mandatory. Formal risk assessment completed before any live test.
9	Sunburn (outdoor testing)	Safety	H	L	M	Sunscreen applied; outdoor sessions scheduled before 10 am or after 2 pm where practical.
10	Fire risk	Safety	L	C	H	Fire extinguisher present in lab. Rated fuses and current limits enforced. Energised hardware never left unattended.
11	Assignment deadline collision	Project	H	M	H	One-month contingency buffer in schedule. Milestones reviewed weekly. Non-critical tasks deprioritised during crunch periods.
12	Skill gap between team members	Team	M	M	M	Work division follows domain expertise (control vs. power electronics). Knowledge transfer at design review milestones.
13	Control–hardware integration failure	Control Sys.	M	H	H	QSpice HIL validates firmware–hardware interaction. Subsystem verification stage isolates faults before full system test.
14	MPPT fails to track global MPP	Control Sys.	M	H	H	Global MPP tracking is a hard requirement. Algorithm validated across all partial shading scenarios in simulation before deployment.
15	Intellectual property dispute	Legal	L	H	M	Monitor IP Australia. All design decisions timestamped in GitHub. Novel details withheld from publication prior to any IP filing.
16	Poor commercial adoption	Strategic	L	L	L	Outside capstone scope. Universal application requirement prevents overfitting to a single use case.
17	Data breach / cybersecurity	Data	L	M	L	No sensitive personal data collected. Repositories private. Credentials excluded from version control.
18	Team member illness	Team	M	M	M	Buffer absorbs short-term absence. Control systems work proceeds in isolation if hardware team is unavailable. Documentation kept current for handover.

Table 5: System Requirements

Category	Requirement
Power Electronics	
Overload protection	The shuffler must incorporate overload protection to prevent damage under fault conditions.
Bidirectional flow	The shuffler must support bidirectional power flow between adjacent PV modules.
Power rating	Rated to the highest commercially available solar panel specifications.
Isolation	Non-isolated topology (galvanic isolation not required).
Switching speed	Minimum switching frequency of 10 kHz.
Partial Shading Validation Scenarios	
Whole panel	Complete darkness or total shading of one or more panels.
Half-panel	Partial shading from a large obstruction covering approximately half a module.
Single string	One complete string fully shaded.
Multiple modules	Multiple non-adjacent modules within a string shaded simultaneously.
Single module	One module within a string shaded.
Control Systems	
Convergence speed	Fast convergence to the MPP — tracking speed is a primary performance metric.
Global MPP tracking	Must track the global MPP and not converge to a local maximum under partial shading.
Individual control	Each shuffler must be independently controllable.
String-level control	Groups of shufflers within a single string must be collectively controllable.
Multi-string control	Groups of shufflers across multiple strings must be collectively controllable.
MPPT allocation	One MPPT tracker per string.
Sensor filtering	Sensor noise must be filtered prior to use in the control loop.
Safety-critical code	Firmware must comply with MISRA-C for safety-critical embedded systems.
Shuffler Design Constraints	
Converter topology	Designed for a typical MPPT system; boost converter topology assumed.
Load assumption	Designed for energy storage system loads only — voltage sources and/or resistive loads.
Application scope	Universal commercial application — not targeted at a single installation type or use case.

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